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Enhanced volatile fatty acids production during anaerobic digestion of lignocellulosic biomass via micro-oxygenation

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HIGHLIGHTS

- The effects of micro-oxygenation on volatile fatty acids (VFAs) production during anaerobic digestion was studied.
- Inoculum type, incubation time, and oxygen dosage showed significant interactions.
- Anaerobically digested cattle manure (ADCM) was the ideal inoculum for VFAs production.
- VFAs yield showed a quadratic correlation with oxygen dosage.

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ABSTRACT

A series of batch experiments were conducted to investigate the effects of inoculum type, oxygen (O_2) dosage, and incubation time on volatile fatty acids (VFAs) production during anaerobic digestion (AD) of Napier grass (*Pennisetum purpureum*), a high yielding energy crop. The results showed that anaerobically digested cattle manure (ADCM) as an inoculum generated significantly higher VFAs compared to that of anaerobically digested waste activated sludge (ADWAS) as an inoculum. Additionally, the incubation time of 3 days and O_2 dosage of 15 mL/g volatile $solids_{added}$ showed the highest VFAs production when ADCM was used as an inoculum. Moreover, the VFAs production had a quadratic correlation with O_2 dosage with R^2 of 0.86. The Scanning Electron Microscopy (SEM) images of the digested fiber showed rough and crumbled surface structures as opposed to that of the undigested fiber, which was further confirmed by changes in structural composition of the digested fiber.

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1. Introduction

Anaerobic digestion (AD) has been widely adopted for bioenergy production from different organic feedstocks, such as forestry and agricultural residues, animal manures, organic fractions of municipal solid wastes, food wastes, and energy crops among others. AD has several merits such as generating renewable energy, remediating wastes, mitigating greenhouse gas emissions and providing energy security. In 2013, there were over 14,500 commercial AD plants in operation in Europe alone (EBA, 2014). The AD process primarily converts organic feedstocks into a single product, i.e., biogas. For complex substrates such as lignocellulosic biomass, the conventional AD process does not maximize the substrate conversion into biogas (Kaparaju et al., 2009; Sawatdeenarunat et al., 2015). One approach to maximize the substrate utilization efficiency could be via an AD-based biorefinery, in which the substrate can be converted into multiple products, such as biogas, volatile fatty acids (VFAs), mixed alcohols, organic fertilizer, etc., (Sawatdeenarunat et al., 2016). VFAs are intermediate products produced during the acidogenic step of the AD process, and are precursors to a plethora of products such as biogas via methanogenesis, alcohol-based fuels (e.g., ethanol and butanol) via carboxylic acid platform (Agler et al., 2011), and biodegradable plastic (i.e., polyhydroxyalkanoates (PHAs)) via bio-polymerization (Yu, 2001). VFAs, especially acetic acid, produced during AD, is an initial substrate for producing polyvinyl acetate (PVA), which is an adhesive for wood, paper, textile, and ceramic. Moreover, PVA is an environmentally friendly polymer and has been approved to be used in food and pharmaceutical packaging (Chaabouni and Boufi, 2017). Moreover, VFAs can be used as an alternative to the organic carbon source in biological nutrient removal (Lee et al., 2014). VFAs can also be used to grow a lipid-accumulating oleaginous yeast (i.e., *Yarrowia lipolytica*) which can further be used as a feedstock for biodiesel production (Fontanille et al., 2012). VFAs

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production using mixed microbial cultures, however, requires suppression of VFAs consumers, especially methanogens. Several methods have been applied to suppress or eliminate methanogenesis. Chemicals such as 2-bromoethanesulfonate (BES) and iodopropane have been widely used as methanogens inhibitors (Chae et al., 2010; Zhu and Béland, 2006). However, the continual exposure of these chemicals to the methanogens resulted in their adaption to the chemicals thereby requiring higher dosage of chemicals during long-term operation. The cost of chemicals and their residual concentration in the digestate could be an impediment for a large-scale VFAs production. Heat-shock and acid treatment of inoculum were also reported to suppress methanogenic activity (Agler et al., 2011; Khanal et al., 2006; Zhu and Béland, 2006). Such approaches also suffer from scalability and operating cost. Thus, there is a need of a benign yet scalable and low cost approach for effective control of methanogenic activity to maximize VFAs yield. Micro-oxygenation is one of the most environmentally friendly methods that inhibits obligate anaerobes such as methanogens (Chae et al., 2010). Fortuitously, micro-oxygenation also facilitates the growth of cellulolytic microbes that produce hydrolytic enzymes thereby improving hydrolysis (Botheju and Bakke, 2011). Several studies reported the positive effects of micro-oxygenation on hydrolysis of various substrates including primary sewage sludge (Johansen and Bakke, 2006), fresh market wastes (Nguyen et al., 2007), brown water and food wastes (Lim and Wang, 2013), and vegetable and flower wastes (Zhu et al., 2009). Lim et al. (2014) reported more diverse bacterial communities in the AD system with micro-aeration than for non-aerated conditions. The authors further observed that, compared to an anaerobic system, a micro-aerated AD system contained a higher proportion of populations belonging to the *Firmicutes* phylum, which resulted in a higher substrate hydrolysis rate. In another study, Lim and Wang (2013) demonstrated the positive effects of micro-aeration on easily degradable substrates (i.e., brown water and food wastes) with increase in VFAs production by nearly 330% after 4 days of micro-oxygenation. The positive effects might be due to the increased activity of facultative hydrolytic and acidogenic microorganisms under the micro-aerobic condition. Zhu et al. (2009) also reported that the efficiency of acidogenesis depends on the degree of oxygenation. Thus, the synergistic effect of inhibition of methanogens and enhancement of facultative microorganisms by micro-oxygenation could effectively increase VFAs production. In this study, a high-yielding energy crop (Napier grass) was used as a lignocellulosic feedstock for VFAs production. Napier grass (*Pennisetum purpureum*) is a perennial herbaceous crop which is native to Africa, and can be cultivated in other regions with a tropical or sub-tropical climate including Hawai'i (Takara and Khanal, 2011). This crop, typically used as a forage for livestock, could be a promising feedstock for second-generation bioenergy and bio-based products generation because of its high biomass yield (Zewdu et al., 2003). Osgood et al. (1996) reported an annual yield of 94 dry metric tons per hectare from ratoons, when cultivated on the Hawaiian island of Molokai. The reported yield is more than double the biomass yield of switch grass (*Panicum virgatum*) which is a common energy crop for biofuel production in the continental United States. (McLaughlin and Adams Kszos, 2005; Takara and Khanal, 2011). The annual methane production potential of Napier grass is estimated to be approximately 8000 m^3/acre (Osgood et al., 1996; Surendra and Khanal, 2014). Although micro-oxygenation has been reported to enhance VFAs production during AD of several substrates, there is a lack of studies using lignocellulosic biomass, especially Napier grass. Thus, the overall goal of this study was to examine the feasibility of micro-oxygenation for enhanced VFAs production using lignocellulosic biomass and to evaluate the interactive effects, if any, of oxygen dosage, inoculum type, and incubation time on VFAs yield from Napier grass.

2. Methods

2.1. Substrate

Napier grass was cultivated for three months and harvested from the University of Hawai'i's Waimanalo Research Station (Waimanalo, HI, USA). The hand-harvested biomass was shredded using a commercial cutting mill (Vincent Corporation, Tampa, FL, USA) to a size of about 2 cm. Later, it was air-dried to a total solids (TS) content of over 90%, to facilitate storage at room temperature without degradation. The dried biomass was then passed through a laboratory cutting mill (Retch SM2000, Haan, Germany) with a screen size of 6 mm. The milled biomass was then stored in vacuum bags for further analysis and was used as a feedstock for the entire experiments. The TS, volatile solids (VS), and fiber composition including Neutral Detergent Fiber (NDF), Acid Detergent Fiber (ADF), and Acid Detergent Lignin (ADL) of the prepared biomass were also analyzed using the methods discussed in the latter sections.

2.2. Inoculum

Two different inocula, namely anaerobically digested cattle manure (ADCM) and anaerobically digested waste activated sludge (ADWAS), were used to investigate the effect of inocula on VFAs yield. ADWAS was collected from an anaerobic digester treating waste activated sludge (Hawai'i Kai, Honolulu, USA) at mesophilic conditions. Similarly, ADCM was taken from a 20-L mother reactor fed with cattle manure and operated at mesophilic conditions. The reactor contents were withdrawn from the digester after 42 days of digestion and sieved using a #8 sieve (ASTM 2.36 mm, Thermo Fisher Scientific Inc., USA) with opening of 2.38 mm to remove the fibers, which could interfere with the compositions of experimental samples. The filtrate after sieving was used as an inoculum for the experiments. The TS and VS contents of the ADWAS were 2.96 ± 0.04% and 2.00 ± 0.04%, respectively. Corresponding TS and VS contents of the ADCM were 3.63 ± 0.35% and 2.71 ± 0.26%, respectively. The prepared inocula were purged with nitrogen gas and stored at 4 °C under anaerobic condition and were reactivated for 3 days at 37 ± 1 °C, before being used in the batch experiments.

2.3. Experimental design and statistical analyses

The experiments were performed following a Randomized Complete Block Design (RCBD). Three factors were considered in the RCBD including inoculum source, O_2 dosage, and incubation time. The inoculum source had two levels, while the O_2 dosage and incubation time had three levels as shown in Table 1. In total, 18 different treatment combinations were tested in each replication. Analysis of Variance (ANOVA) with a threshold value (α) of 0.05 followed by a post hoc Tukey's test of experimental data was conducted. Biogas and VFA yields were used as response variables for the different treatment conditions. All the statistical analyses were performed using JMP statistical software (JMP Pro 12.0.1, SAS Institute Inc., Cary, NC, USA).

2.4. Batch experimental setup

A series of batch experiments were conducted using a 250-mL Erlenmeyer flasks with a working volume of 160 mL. The dried and milled Napier grass was used as the substrate. A substrate-to-inoculum ratio was maintained at 1:1 based on VS. TS content in each flask was adjusted to 4% using distilled water. Controls were set up in the same way as the samples except Napier grass

Table 1
Experimental design pattern.

Factor	Type	Levels	Types or values	Unit
Inoculum source	Fixed	2	ADCM, ADWAS	–
Oxygen dosage	Fixed	3	0, 10, 15	mL/gVS$_{added}$
Incubation time	Fixed	3	1, 3, 5	day(s)

to measure VFAs and biogas production from inocula alone during the experiments. The flasks were then injected with O_2 following the experimental design just prior to the start of the incubation. All the experiments were conducted in an incubator shaker (New Brunswick Scientific Excella™ E25, New Brunswick Scientific Co., Inc., USA), where the temperature and shaking speed were maintained at 37 ± 1 °C and 100 rpm, respectively. The experiments were terminated at different time periods based on the experimental design. After each incubation period, the content from each flask was sieved through a 0.85-mm screen. The filtrate of the samples following sieving was centrifuged and subsequently analyzed for individual VFAs. The produced biogas was quantified and the composition was analyzed. Based on the results from the 250-mL flask experiments, the conditions that resulted in the highest, medium, and the lowest VFAs for each inoculum type, were retested in a series of 2-L bottles with a working volume of 1.5 L to collect enough digested biomass samples for composition analysis of fiber. The selected conditions were significantly different from each other. All the experiments were conducted in triplicates to confirm the repeatability and reproducibility of the results. The statistical correlation between VFAs production and the O_2 dosage was also evaluated. The physical morphology of digestate fiber was compared with untreated biomass samples using Scanning Electron Microscopy (SEM) to examine the structural changes that occurred during AD.

2.5. Optimization studies and experimental verification

The results obtained from the batch experiments with different O_2 dosages, incubation times and inoculum types were modeled using a quadratic regression. The first-degree derivative of the equation from the quadratic regression was performed with respect to O_2 dosage and was set to zero to predict the O_2 dosage which gives the maximum VFAs production as illustrated in Eq. (1):

$$Y_{VFAs} = AX_{ox}^2 + BX_{ox} + C \quad (1)$$

where, Y_{VFAs} is VFAs yield (mg/gVS$_{added}$), X_{ox} is O_2 dosage (mL/gVS$_{added}$), and A, B and C are regression coefficients.

A series of batch tests were then reconducted in a 250-mL Erlenmeyer flask to test the prediction model using the predicted O_2 dosage obtained from the model that resulted in the maximum VFAs yield. The experiment was conducted in triplicate. Furthermore, the predicted and experimental values were statistically compared.

2.6. Analytical methods

pH was measured using a bench top pH meter (Accumet AB15, Fisher, Fairlawn, USA). TS and VS were analyzed following the Standard Methods (APHA et al., 2005). Fiber compositions (i.e., NDF, ADF, and ADL) were analyzed before and after AD using a cell wall fractionation method, according to Van Soest (Faithfull, 2002). Biogas production was quantified using a milligas counter (Ritter US LLC. NY, USA), which works based on the principle of buoyancy. The biogas compositions, including methane and carbon dioxide, were analyzed using gas chromatography equipped with thermal conductivity detector (GC-TCD) (GC2014, Shimadzu, Japan) with a packed column (80/100 Hayesep D column, 2 m length × 3.2 mm outer diameter × 2.1 mm inner diameter, Supelco, USA). The individual VFAs (i.e., acetic, propionic, isobutyric, butyric, isovaleric and valeric acids) were quantified using GC equipped with flame ionization detector (GC-FID) (GC2014, Shimadzu, Japan) with a capillary column (ZB-Wax Plus column 30 m length × 0.25 mm inner diameter × 0.25 μm film thickness, Phenomenex, USA). For Scanning Electron Microscopy (SEM) analysis, the fiber specimens were mounted on a conductive carbon tape with aluminum stubs and the sputter was coated with gold/palladium in a Hummer 6.2 sputter coater. Then the specimens were viewed with a field emission SEM (Hitachi S-4800, japan) at an accelerating voltage of 5.0 kV.

3. Results and discussion

AD of Napier grass for VFAs production was evaluated in batch-mode under micro-oxygenation condition. A series of experiments were conducted based on a RCBD to investigate the VFAs yield using two different inocula, and at three different O_2 dosages and three different incubation periods. The major findings of this study are discussed in the following sections.

3.1. Effect of inoculum on VFAs yield

ADCM as an inoculum produced significantly ($\alpha = 0.05$) higher total VFAs under different O_2 dosages and incubation times as shown in Fig. 1 and Table S1 in Supplementary Information (SI). The lowest VFAs yield from ADCM batch at O_2 dosage of 30 mL/gVS$_{added}$ and incubation time of 1 day was twofold higher than the highest VFAs yield when ADWAS was used as an inoculum under similar conditions. Overall, the batch tests using ADCM as an inoculum resulted in nearly 13-fold higher VFAs yield compared with the tests using ADWAS as an inoculum during the incubation time of 3 days. In general, cattle manure have phylogenetically diverse microbial communities including many cellulolytic bacteria that release hydrolytic enzymes capable of enhancing hydroly-

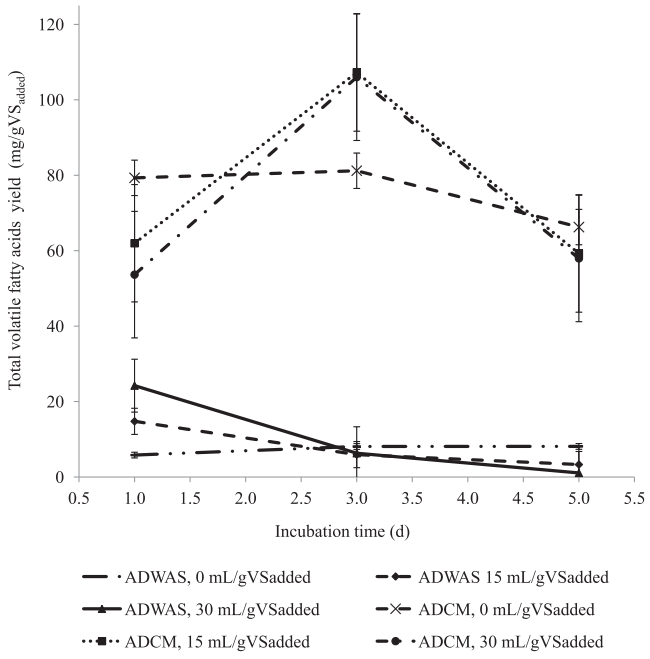


Fig. 1. Total volatile fatty acids yield from Napier grass.

sis of lignocellulosic biomass (Gupta et al., 2016). Thus, ADCM inoculum plays key role in enhanced VFAs production from lignocellulosic feedstocks. Similar results were also reported by Gu et al. (2014) when digested dairy manure and digested municipal sludge were compared as inocula for AD of lignocellulosic biomass (i.e., rice straw). The study showed an 8-fold higher specific methane yield with a shorter lag phase when the digested cattle manure was used as an inoculum compared to digested municipal sludge as an inoculum. The higher enzyme activities (i.e., cellulases and xylanases) and the availability of suitable nutrients in anaerobically digested dairy manure were the likely contributor to the higher VFAs yield with ADCM inoculum over ADWAS in this study (Gu et al., 2014). Surendra and Khanal (2014) also reported higher methane yield when ADCM was as an inoculum for AD of Napier grass in a series of batch studies to examine biomethane production potential. In the United States alone, over 120 million dry tons of cattle manure is produced annually (Yue et al., 2010), which could be used in co-fermentation with lignocellulosic feedstocks for VFAs production in full-scale AD plants. This apparently suggests that cattle manure-derived inoculum is ideal for VFAs production from lignocellulosic feedstocks

3.2. Effect of oxygen dosage and incubation time on VFAs yield

O_2 dosage by itself did not show a significant difference in VFAs yield between the two inocula. However, incubation time as well as the interaction between O_2 dosage and incubation time showed a significant difference. This could be attributed to an increase in the population of facultative microorganisms (Botheju and Bakke, 2011) and the enhancement of hydrolytic extracellular enzymes produced (Johansen and Bakke, 2006) during micro-oxygenation. The incubation time is also an important factor, which could affect VFAs yield and composition (Bengtsson et al., 2008). Zhu et al. (2009) investigated the importance of incubation time for microorganisms to acclimate and adapt to the AD's operating conditions under micro-oxygenation in the leach-bed reactor using fresh vegetable and flower wastes as the substrates without inoculation. The VFAs production in the micro-oxygenated reactor increased from day 2 to day 5 and then started to decrease. However, in this study, the VFAs yield of batch tests with ADCM as inoculum reached a plateau on day 3, even though Napier grass has a higher lignin (ADL) content compared to the mixture of flower and vegetable wastes (i.e.,10.2% and 2.3% of TS, respectively). The enhanced and sustained VFAs yield at a shorter incubation time in this study could be attributed to a well-acclimated inoculum used in this study. Moreover, Zhu et al. (2009) also reported that VFAs produced during oxygenation is dependent on the amount of O_2 input. The authors further reported that insufficient micro-aeration could negatively affect the development of facultative microorganisms thereby resulting in poor system performance. Contrary to this, an appropriate micro-aeration rate could promote the hydrolysis process and result in higher VFAs yield. Jagadabhi et al. (2010) reported over 4-fold increase in VFAs production during AD of grass silage using leach-bed reactors, when aeration was supplied at a flow rate of 1 L/min compared to non-oxygenated conditions. As presented in Fig. 2, with batch tests using ADCM as inoculum, the average daily methane yield between day 1 and day 3 of incubation period was 10.3 ± 3.0 mL/gVS$_{added}$-d, which was less than half of the yield between day 3 and day 5. Thus, it is apparent that methanogens started to grow during longer incubation time.

On the other hand, the batch tests using ADWAS as an inoculum had the average daily methane yield of 29.8 ± 2.6 mL/gVS$_{added}$-d between day 1 and day 3 of incubation period which was higher than the methane yield of 16.5 ± 5.9 mL/gVS$_{added}$-d for the incubation period between day 3 and day 5. Thus, the use of ADWAS resulted in higher conversion of VFAs into methane in the first

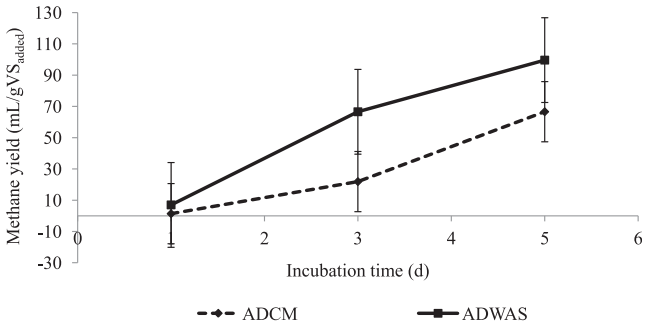


Fig. 2. Methane yield at various incubation times.

3 days of incubation period. The highest VFAs yield observed in this study was 107.3 ± 2.6 mg/gVS$_{added}$ at O_2 dosage of 15 mL/gVS$_{added}$, incubation time of 3 days and ADCM as an inoculum. The VFAs yield was not significantly different from that at O_2 dosage of 30 mL/gVS$_{added}$ and incubation time of 3 days. Thus, micro-oxygenation can be used as an environmentally friendly approach of inhibiting methanogenesis for enhanced VFAs production.

The highest VFAs yield obtained in this study (i.e., 107.25 ± 2.19 mg/gVS$_{added}$) was lower than that reported by Jagadabhi et al. (2010) (i.e., 139.50 mgVFAs/gVS$_{added,}$) when grass silage was used as the substrate. The higher yield of VFAs from the grass silage compared to raw Napier grass used in this study was attributed to ensiling, which often serve as a biological pretreatment at acidic conditions (i.e., pH 4.0 to 4.5), which further enhanced the hydrolysis during AD.

3.3. VFAs composition

As shown in Fig. 3, the use of ADCM as an inoculum generated a more diverse variety of individual VFAs compared to ADWAS as an inoculum. For both inocula, acetic and propionic acids dominated the VFAs profile. However, with ADWAS as an inoculum, acetic and propionic acids were the major individual VFAs accounting for 50 to 100% of the total VFAs. Zhu et al. (2009) also reported a similar pattern with the dominance of acetic, propionic, and butyric acids during AD of fresh vegetable and flower wastes under sufficient micro-aerated conditions. Similarly, acetic and butyric acids were the major VFAs produced during AD of grass silage (Jagadabhi

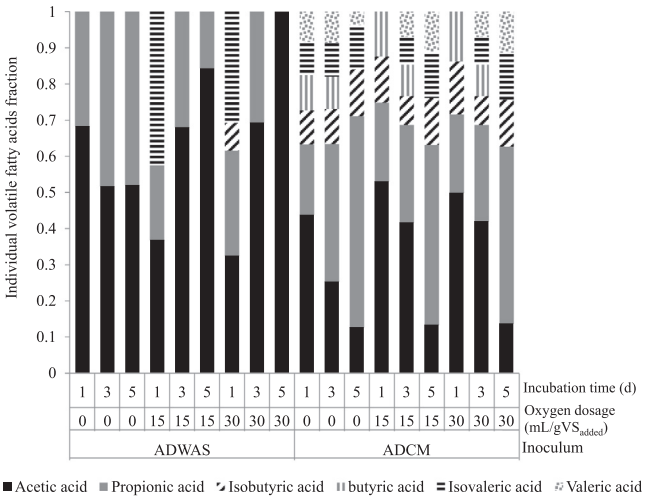


Fig. 3. The distribution of individual VFAs.

et al., 2010). With ADWAS as an inoculum, acetic acid concentration increased with increasing incubation time and contributed over 80% of VFAs at an incubation time of 5 days in the micro-oxygenated batches. This was mainly due to conversion of higher carbon VFAs into acetic acid by the acclimated acetogenic microorganisms in ADWAS. Under non-oxygenated conditions, the produced acetic acid was not significantly different. Conversely, the amount of acetic acid produced with ADCM as an inoculum decreased at longer incubation time. Acetic acid, which is the only VFAs that methanogens can consume, was converted into methane during long incubation time, which facilitated the growth of methanogens as evident from the increasing methane yield as shown in Fig. 2. Thus, the distribution of individual VFAs strongly depended on the types of inoculum used.

3.4. Statistical prediction model for VFAs yield

As seen from Fig. 1, the optimum condition for high VFAs yield was obtained when ADCM was used as an inoculum for the incubation time of 3 days. The correlation between VFAs yield and O_2 dosage under the selected condition was further studied by applying quadratic regression model. The VFAs yield showed a close fit to the quadratic model with the coefficient of determination (R^2) of 0.86. The predicted model was found to be $Y_{VFAs} = -0.06X^2_{ox} + 2.65X_{ox} + 81.20$, where Y_{VFAs} and X_{ox} represent VFAs yield and O_2 dosage, respectively. The optimum O_2 dosage was then calculated from the first-order differentiation of the equation. The predicted optimum O_2 dosage in this study was found to be 22 mL/g VS_{added}, which resulted in the maximum VFAs yield of 110.05 ± 5.46 mg/g VS_{added}. To further prove that the predicted value truly reflects the actual value, a series of batch experiments was conducted in triplicate using the O_2 dosage of 22 mL/g VS_{added}, incubation time of 3 days, and ADCM as inoculum. The experimental results showed that the VFAs yield of 112.70 ± 5.15 mg/gVS_{added}, was not significantly different from the predicted value (i.e., 110.05 ± 5.46 mg/gVS_{added}) obtained using above equation. Thus, the proposed model was precise enough to effectively predict the VFAs yield during AD of Napier grass using ADCM as inoculum during incubation time of 3 days. Since using ADCM as inoculum and incubation time of 3 days was the optimum operating condition in this study, the constructed model could effectively predict the required O_2 dosage to obtain the expected VFAs production without performing additional experiment.

3.5. Changes in structural composition of biomass

The best, the median, and the worst operating conditions for VFAs productions using both ADCM and ADWAS inocula were selected (Table 2) to investigate the changes in structural composition of biomass during AD.

The structural carbohydrates (i.e., cellulose and hemicellulose) and lignin (ADL) contents of Napier grass before and after AD are summarized in Table 3. Compared with raw Napier grass, the fibers obtained from the batch tests with ADCM as inoculum showed significantly lower amount of hemicellulose (i.e., 25–34%) and cellulose (i.e., 2–18%). Conversely, for the fibers obtained from batch tests using ADWAS as inoculum, there were no significant difference in both hemicellulose and cellulose contents in the biomass before and after AD (Table 3 and Table S2 in SI). The higher cellulose and hemicellulose degradation of fibers obtained from the batch tests using ADCM as inoculum thus resulted significantly higher VFAs yields compared with the batch tests using ADWAS as an inoculum. Gu et al. (2014) also presented a similar discussion when digested dairy manure and digested municipal sludge, and rice straw were used as the inoculum and substrate, respectively for methane production. The authors reported an increase in cellulose and hemicellulose degradation resulting in higher specific methane yield. For the batch tests using ADWAS as an inoculum, there were no significant differences in biomass composition after AD among all operating conditions. This further showed that ADWAS was the least effective inoculum in the AD of Napier grass. Surendra and Khanal (2014) also observed high methane yield from Napier grass, when ADCM was used as an inoculum. Typically, the macromolecules, such as lignocellulose are degraded during hydrolysis step of AD to produce monomeric sugars, which are subsequently converted into VFAs and then to methane. Thus, the higher the rate of degradation of these compounds, the higher the product yield (i.e., VFAs and/or methane) would be. Based on the compositional analysis, it was apparent that the microorganisms preferably consumed hemicellulose while leaving cellulose and lignin in the digestate during AD. Approximately 25–34% and 21–23% of hemicellulose were degraded from the raw Napier grass using ADCM and ADWAS as inoculum, respectively. However, the respective cellulose degradations were only 2–18% and 9–18% of total cellulose using ADCM and ADWAS as inoculum. Yue et al. (2010) studied the composition change of cow manure fiber during AD and reported that hemicellulose was the favorable component of

Table 2
The selected conditions to examine the changes in structural composition of biomass.

Operating condition	Inoculum			
	ADCM		ADWAS	
	Oxygen dosage (mL/gVS_{added})	Incubation time (day)	Oxygen dosage (mL/gVS_{added})	Incubation time (day)
Best	15	3	30	1
Median	0	1	30	3
Worse	30	1	30	5

Table 3
The characteristics and structural carbohydrates under the selected conditions.

Sample	NDF (% TS)	ADF (% TS)	Lignin (ADL) (% TS)	Hemicellulose (% TS)	Cellulose (% TS)
Raw Napier grass	75.0 ± 1.0	49.7 ± 0.6	10.2 ± 0.3	25.3 ± 1.6	39.5 ± 0.9
ADCM,15,3	73.4 ± 0.6	55.4 ± 1.8	23.0 ± 1.0	19.0 ± 1.3	32.3 ± 1.0
ADCM,0,1	73.0 ± 0.6	55.3 ± 2.9	18.9 ± 0.6	17.7 ± 2.2	36.3 ± 0.6
ADCM,30,1	73.3 ± 1.1	56.5 ± 0.9	17.8 ± 2.4	16.8 ± 1.0	38.6 ± 2.4
ADWAS,30,1	66.6 ± 2.4	46.7 ± 0.2	11.0 ± 1.3	19.9 ± 1.4	35.7 ± 1.3
ADWAS,30,3	69.2 ± 4.0	49.6 ± 0.4	13.7 ± 4.7	19.5 ± 1.4	36.0 ± 4.7
ADWAS,30,5	66.7 ± 1.5	46.8 ± 0.2	14.3 ± 3.1	19.9 ± 2.8	32.5 ± 3.1

Based on n = 6 for each sample.

the fiber to be converted to methane. The digested fiber contained 11% less hemicellulose compared with raw manure (Yue et al., 2010). The cellulose-rich fiber has undergone partial biological pretreatment during AD and could directly be subjected to enzymatic hydrolysis to release monomeric sugars as a-potential feedstock for producing other bio-based chemicals via anaerobic biorefinery approach (Sawatdeenarunat et al., 2016). Ethanol is one of such products that could be obtained from cellulose-rich fiber. The ethanol production from low-hemicellulose feedstock could avoid the problem of fermenting five carbon sugar into ethanol, which has low ethanol yield and requires specific microorganisms (Lee, 1997).

The lignin content measured as ADL increased for all experimental conditions (i.e., 75–125% for biomass samples when ADCM was used as inoculum and only 8–40% for biomass samples when ADWAS was used as inoculum) following AD compared to the raw Napier grass due to decrease in cellulose and hemicellulose contents of the fiber. Lignin, a phenylpropane-based polymer, is highly recalcitrant to microbial attack including AD. Since the ANKOM method was used for fiber analysis, which is a gravimetric mass balance approach, residual fibers showed higher lignin content. Lignin can also be used as a substrate to produce many bio-based products, i.e., vanillin (a flavoring reagent), a binding agent in an animal feed, carbon fiber as well as heat and electricity via thermochemical processes (Surendra et al., 2015).

3.6. Microscopy examination of biomass

Digested fiber samples were collected from the serum bottles that yielded the highest VFAs, and were subjected to Scanning Electron Microscopy (SEM) examination. The SEM micrographs of the digested fiber showed rough and crumbled surface structures as opposed to the undigested fiber, which had smooth intact surface structures. The rough surface of the fiber was mainly due to degradation of hemicellulose from the biomass as evident from the fiber analysis presented in Table 3. The surface structure showed some correlation between the fiber destruction and VFAs yield. It is important to note that the rough surface structure does not necessarily correlate to VFAs yield as the fibers collected from the serum bottles at O_2 dosage of 30 mL/gVS_{added} and incubation time of 1 day did not yield significantly higher VFAs yield. Thus, additional parameters such as the composition of the raw and digested fiber need to be examined as discussed in Section 3.4.

4. Conclusion

VFAs production from Napier grass was examined using different inocula, micro-oxygenation dosages, and incubation times. Using anaerobically digested cattle manure as an inoculum showed significantly higher VFAs yield than the anaerobically digested waste activated sludge as an inoculum. There was significant interaction between micro-oxygenation and incubation time, which played important role to enhance VFAs yields from Napier grass. Thus, micro-oxygenation could be an environmental-friendly strategy to enhance VFAs yields from lignocellulosic feedstocks for high-value chemicals and bioenergy production.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.biortech.2017.02.029.

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